The exact critical scaling of the uniform doubling condition with truncations

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Abstract

This note completes the determination of the critical scaling of the uniform doubling condition of [Shmalo, *Ann. Math. Artif. Intell.* (2024)] for independent samples, begun in three companion papers. There we showed: the condition holds with probability $1 - 1/N$ with scaling factor $\ell_N = O((\log N/N)^{1/n})$ for all uniformly thick truncated-slab families; for pure slabs over smooth bodies the critical scaling is exactly $\Theta((\log N/N)^{2/(n+1)})$; and a single conical truncation forces $\ell \geq cN^{-1/n}$ with probability bounded below. Here we close the last gap: for an explicit family of uniformly thick conical truncations, with probability $1 - o(1)$ the condition fails at every grid scale below $c\,(\log N/N)^{1/n}$. Hence the truncated critical scaling is $\Theta\big((\log N/N)^{1/n}\big)$, exactly, logarithm included, and the two-regime picture is complete: $\Theta((\log N/N)^{2/(n+1)})$ without truncations on smooth supports, $\Theta((\log N/N)^{1/n})$ with them. The proof is a second-moment argument over $\asymp N/\log N$ translated cones, whose apex caps are disjoint even though the cone bodies overlap.

1 Setting and result

Let $K = [0,1]^n$, $n \geq 1$, let μ be the uniform measure on K, and let $T = \{X_1, \dots, X_N\}$ be i.i.d. samples with empirical measure $\bar{\mu}_N$. Truncated slabs $S_w = \{x : |\langle x - s, u\rangle| \leq \frac{w}{2}\} \cap H$ and the uniform doubling condition (UDC) with constants $(\kappa, \delta, \sigma, \beta, \ell)$ and truncation family $\mathcal{L}$ are as in [1, 2]: for every truncated slab generated by $\mathcal{L}$ and every grid scale $\ell\kappa^i < \beta$, $\bar{\mu}_N(S_{\ell\kappa^{i+1}}) \geq \min\{\delta, (1+\sigma)\bar{\mu}_N(S_{\ell\kappa^i})\}$. By [2], the UDC holds with probability $1 - \frac{1}{N}$ and scaling factor $\ell_N = O((\log N/N)^{1/n})$ simultaneously for all families of truncation sets of bounded size whose nonempty regions contain a ball of fixed radius r_0. By [3], a single thick conical truncation forces $\ell \geq cN^{-1/n}$ with probability bounded below by a constant, and by [4] the pure-slab critical scaling on smooth bodies is exactly $\Theta((\log N/N)^{2/(n+1)})$. We prove:

Theorem 1.1. *Fix $n \geq 1$, $\kappa > 1$, $\sigma > 0$, $\beta > 0$ and $\delta \geq \frac{3}{N}$. There are a constant $r_0(n) > 0$, an explicit family $\mathcal{L}_N$ of truncation sets of size n whose regions all contain a ball of radius $r_0(n)$, and constants $\varepsilon_1(n,\kappa) > 0$, $c(n,\kappa) > 0$, such that with probability $1 - o(1)$ the doubling inequality fails, for some slab truncated by a member of $\mathcal{L}_N$, at every grid scale below $c\,(\log N/N)^{1/n}$. Consequently, with probability $1 - o(1)$, the UDC for the family $\mathcal{L}_N$ holds with no scaling factor*

$$\ell \;\leq\; c\Big(\frac{\log N}{N}\Big)^{1/n}.$$

Together with the upper bound of [2], the critical scaling of the uniform doubling condition with (even a single size class of) truncations is $\Theta\big((\log N/N)^{1/n}\big)$: exponent and logarithm are both exact.

We work in the cube and with orthant-shaped (size-n) truncation sets for explicitness; the construction adapts to any support with density bounded above and below, and to single-half-space truncations of smooth bodies as in [4], with the same exponent.

2 The cone family and the stall events

Write $e := \mathbf{1}/\sqrt{n}$ and $\tau(x) := \langle x, e\rangle$. For a point $a \in [\frac{1}{4}, \frac{3}{4}]^n$ (the *apex*), consider the truncation set h_a of size n consisting of the half-spaces $\{x : \ x_j \le a_j\}$, $j = 1, \dots, n$, whose region is the rectangular orthant

$$G_a \ := \ \big\{x \in [0,1]^n : \ x_j \le a_j \ \forall j\big\} \ = \ \prod_{j=1}^{n} [0, a_j].$$

Each G_a contains the ball of radius $r_0(n) := \frac{1}{16}$ centered at $a - \frac{\mathbf{1}}{8}$: indeed $B\big(a - \frac{\mathbf{1}}{8}, \frac{1}{16}\big) \subseteq \prod_j \big[a_j - \frac{3}{16}, a_j - \frac{1}{16}\big] \subseteq \prod_j [0, a_j] = G_a$, using $a_j \ge \frac{1}{4} > \frac{3}{16}$. The point a is the τ-maximum of G_a, and for $0 \le h \le \frac{1}{4\sqrt{n}}$ the cap

$$C_a(h) \ := \ \big\{x \in G_a : \ \tau(x) \ge \tau(a) - h\big\}$$

is the corner simplex $\{x \le a, \ \sum_j (a_j - x_j) \le h\sqrt{n}\}$, of volume exactly $\frac{(h\sqrt{n})^n}{n!}$; hence with $\Lambda_0 := \frac{n^{n/2}}{n!}$,

$$\mu\big(C_a(h)\big) \ = \ \Lambda_0 \, h^n, \qquad 0 \le h \le \tfrac{1}{4\sqrt{n}}. \tag{1}$$

Lemma 2.1 (Stall)**.** *Fix $\kappa > 1$, $\sigma > 0$, $\delta \ge \frac{3}{N}$, an apex a, and depths $0 < h \le \frac{H}{4}$ with $H \le \frac{1}{4\sqrt{n}}$. Suppose the sample contains exactly one point of $C_a(H)$, and that point, X^*, lies in $C_a(h)$. Then for every $w \le H/\kappa$, the slab with direction e, center X^* and truncation h_a satisfies*

$$\bar{\mu}_N(S_w) = \bar{\mu}_N(S_{\kappa w}) = \tfrac{1}{N} < \min\big\{\delta, \ (1+\sigma)\,\bar{\mu}_N(S_w)\big\},$$

so the doubling inequality fails at every scale $w \le H/\kappa$; and if $\ell \le H/\kappa^2$ (and $\ell < \beta$), some grid scale lies in $(0, H/\kappa]$ and the UDC with scaling factor ℓ fails.

Proof. For $w' \le H$: $S_{w'} \subseteq G_a \cap \{\tau \ge \tau(X^*) - \frac{w'}{2}\} \subseteq C_a\big(h + \frac{w'}{2}\big) \subseteq C_a(H)$, since $h + \frac{w'}{2} \le \frac{H}{4} + \frac{H}{2} < H$. On the assumed event, $C_a(H)$ contains only X^*, which belongs to every $S_{w'}$; so for $w \le H/\kappa$ both S_w and $S_{\kappa w}$ have empirical mass exactly $\frac{1}{N}$, while $\min\{\delta, (1+\sigma)\frac{1}{N}\} > \frac{1}{N}$ because $\delta \ge \frac{3}{N}$ and $\sigma > 0$. □

3 Proof of Theorem 1.1

Proof. Set $q := (4\kappa^2)^n$, $\varepsilon_1 := \frac{1}{4q}$, $\tau_N := \varepsilon_1 \log N$, and define the inner and outer depths

$$h_N := \Big(\frac{\tau_N}{\Lambda_0 N}\Big)^{1/n}, \qquad H_N := 4\kappa^2 \, h_N,$$

so that by (1), $\mu(C_a(h_N)) = \frac{\tau_N}{N}$ and $\mu(C_a(H_N)) = \frac{q\tau_N}{N}$, for every apex a and all large N (note $H_N \to 0$).

The apex family. All caps $C_a(H)$ satisfy $C_a(H) \subseteq \prod_j [a_j - H\sqrt{n},\ a_j]$, a box of diameter at most $H\sqrt{n}\cdot\sqrt{n} = Hn$. Choose apexes $a_1, \dots, a_M$ on a grid of spacing $2nH_N$ inside $[\frac{1}{4}, \frac{3}{4}]^n$; then the caps $C_{a_j}(H_N)$ are pairwise disjoint and

$$M \;\geq\; \Big(\frac{1}{4nH_N}\Big)^n \;\geq\; \frac{c_1(n,\kappa)\,N}{\tau_N} \tag{2}$$

for large N. Let $\mathcal{L}_N := \{h_{a_1}, \dots, h_{a_M}\}$; every region G_{a_j} contains a ball of radius $r_0(n) = \frac{1}{8}$ as noted above. (The cone *bodies* G_{a_j} overlap heavily; only the caps must be disjoint.)

Per-apex events and first moment. Let

$$E_j := \Big\{ \#\big(T \cap C_{a_j}(H_N)\big) = 1 \text{ and } T \cap C_{a_j}(H_N) \subseteq C_{a_j}(h_N) \Big\}, \qquad X := \sum_{j \leq M} \mathbf{1}_{E_j}.$$

Then, exactly as in [4],

$$p := \mathbb{P}(E_j) = N \cdot \frac{\tau_N}{N}\Big(1 - \frac{q\tau_N}{N}\Big)^{N-1} \;\geq\; \tau_N\, e^{-2q\tau_N} \;=\; \tau_N\, N^{-2q\varepsilon_1} \;=\; \tau_N\, N^{-1/2},$$

for large N, by the choice of ε_1. Hence by (2),

$$\mathbb{E}X \;=\; Mp \;\geq\; c_1\,\frac{N}{\tau_N}\cdot\tau_N N^{-1/2} \;=\; c_1\, N^{1/2} \;\longrightarrow\; \infty.$$

Second moment. For $i \neq j$ the caps are disjoint, so

$$\mathbb{P}(E_i \cap E_j) = N(N-1)\Big(\frac{\tau_N}{N}\Big)^2\Big(1 - \frac{2q\tau_N}{N}\Big)^{N-2}, \qquad \frac{\mathbb{P}(E_i \cap E_j)}{p^2} \;\leq\; \exp\Big(O\Big(\frac{(\log N)^2}{N}\Big)\Big) = 1 + o(1),$$

uniformly in $i \neq j$, since $\log \frac{1-2x}{(1-x)^2} = O(x^2)$ for $0 \leq x \leq \frac{1}{4}$. Hence $\mathrm{Var}(X) \leq \mathbb{E}X + o(1)(\mathbb{E}X)^2$ and $\mathbb{P}(X = 0) \leq \frac{1}{\mathbb{E}X} + o(1) = o(1)$.

Conclusion. With probability $1 - o(1)$ some E_j occurs, and Lemma 2.1 (with $h = h_N$, $H = H_N = 4\kappa^2 h_N \geq 4h_N$) shows the doubling inequality fails for slabs truncated by $h_{a_j} \in \mathcal{L}_N$ at every scale $w \leq H_N/\kappa$. Hence the UDC with the family $\mathcal{L}_N$ requires

$$\ell \;>\; \frac{H_N}{\kappa^2} \;=\; 4h_N \;=\; 4\Big(\frac{\varepsilon_1 \log N}{\Lambda_0 N}\Big)^{1/n} \;\geq\; c(n,\kappa)\Big(\frac{\log N}{N}\Big)^{1/n}.$$

□

Remark 3.1. *The same argument with rotated single-half-space truncations of a smooth body (the corner caps replaced by edge caps) should yield matching logarithmic lower bounds for the intermediate edge-dimension exponents discussed in the endpoint problems of [3]. Together, the four papers determine the complete phase diagram of the uniform doubling condition for i.i.d. data: the critical resolution is the isolation depth of the most isolated extremal sample point of the relevant cap geometry — $\Theta((\log N/N)^{2/(n+1)})$ for smooth supports without truncations, $\Theta((\log N/N)^{1/n})$ in the presence of conical truncation geometry, which is the regime of neural-network activation cells.*

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

References

[1] Y. Shmalo, *Stability of accuracy for the training of DNNs via the uniform doubling condition*, Ann. Math. Artif. Intell. (2024), DOI 10.1007/s10472-023-09919-1.

[2] Y. Shmalo, *The uniform doubling condition holds with high probability for independent samples*, manuscript, 2026.

[3] Y. Shmalo, *The critical scaling of the uniform doubling condition: smooth bodies versus truncations*, manuscript, 2026.

[4] Y. Shmalo, *The logarithm is necessary: exact critical scaling of the uniform doubling condition on the ball*, manuscript, 2026.